MIL-I-52529A(ME)
15 January 1975
SUPERSEDING
MIL-I-52529(ME)
15 September 1966

MILITARY SPECIFICATION

IDLER, CHAIN, DOUBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a double chain idler for 2-inch and 1-inch chain.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-C-650	- Crates, Wood, Open and Covered.

Military

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-P-15328	- Primer (Wash) Pretreatment, Blue (Formula No. 117-B for Metals).

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

FSC 3835

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DRAWINGS

ME

TA13213E8280 - Idler, Double, Chain, Assembly.

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY, INC. (AWS)

D1.1 - Structural Welding Code, Section 5, Qualifications.

(Application for copies should be addressed to the American Welding Society, 2501 North West Seventh Street, Miami, FL 33125.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The idler shall be in accordance with Top Assembly TA13213E8280 and as specified herein.

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3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the idler. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more idlers as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce idlers that comply with the requirements of this specification. Examination and test shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Identification marking. The idler shall be identified in accordance with MIL-STD-130.

3.5 Treatment and painting. Unless otherwise specified herein, the portions of the idler normally painted shall be cleaned, treated and painted in accordance with MIL-T-704, Type B. The surfaces shall be treated as soon as possible after cleaning with an organic pretreatment primer conforming to MIL-P-15328. The finish coat shall be as specified in the drawings. The finish coat, when dry, shall have a smooth, even surface, free from runs, sags or foreign material. Surfaces of components and assemblies not normally painted with a finish coat shall be cleaned and prepared at the supplier's option.

3.6 Lubrication. The double chain idler and all components shall be lubricated on unpainted machined surfaces with grease conforming to MIL-G-10924.

3.7 Workmanship.

3.7.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of

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the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.7.2 Welding. The surfaces to be welded shall be free from rust, scale, paint, grease, loose mill scale, and other foreign matter. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the idler to conform to the examination and test requirements specified in Section 4. Welding shall be done by the shielded electric-arc method. Welds shall transmit stress to the parts connected by the welds without failure or permanent deformation when the idler is loaded with a 20,000-pound load hanging over the 2-inch chain sprocket with the ratchet and dog engaged. Where embrittlement of the joined metal may result, preheating or stress-relieving, or both shall be employed. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.188 inch unless appropriate bridging techniques are used. Welding and the certification of welder operators shall be in accordance with AWS D1.1 Structural Welding Code.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the supplier.

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4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5 and 6.6).

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction idler shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.1.

4.3.2 Test. The preproduction idler shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.1.

4.4 Quality conformance inspection.

4.4.1 Examination. Each idler shall be examined for the major and minor defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Test. Each idler shall be tested as specified in 4.5.2. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. Each idler shall be examined as specified herein for the following defects:

Major

101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.
102. Components missing or not as specified.
103. Any restriction to free rotation of idler wheels.
104. Material not as specified.
105. Workmanship not as specified.

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Minor

201. Any dimension, other than those identified on the SQAP requirements shown on the drawings, not as specified.
202. Identification marking missing, incomplete or not legible.
203. Treatment and painting not as specified.

4.5.2 Test. With the idler rigidly supported in its normal working configuration, and with the ratchet and dog engaged, hand 20,000 pounds over the 2-inch chain sprocket. Nonconformance to 3.7.2 shall constitute failure of this test.

4.6 Inspection of preparation for delivery.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. The preproduction pack shall be examined for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.1.2 Tests. The preproduction pack of the crated idler for Level A shall be subjected to either the pendulum impact test, incline-impact test, or railroad car impact test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The railroad car test is the preferred test and shall be accomplished by humping the car into a string of five empty cars, with draft gear extended and brakes set, at a speed of not less than 10 nor more than 11 miles per hour. Any shifting of contents, loosening or breaking of holddowns, blocking or bracing, or any visible damage to the contents shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

106. Crate not as specified for Level A.
107. Shroud not as specified for Level A.
108. Contents not anchored as specified for Level A.
109. Marking illegible, incomplete or incorrect.

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5. PREPARATION FOR DELIVERY

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the supplier shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packing that the packing and marking comply with the preparation for delivery requirements of this specification. Examination and tests shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The preproduction pack may be accomplished utilizing either the preproduction model idler or production idler. When the preproduction model is utilized, any packing shall be removed by the supplier at no expense to the Government, when requested by the Government, to facilitate comparison between the preproduction model and production idlers.

5.2 Packing. Packing shall be Level A or C, as specified (see 6.2).

5.2.1 Level A. The idler shall be packed in a crate conforming to PPP-C-650, Type V, Style B, open. The crate shall be provided with a shroud as specified in the crate specification appendix. The windlass keeper shall be secured with wire to prevent loosening. Anchoring of the crate contents shall be in accordance with the appendix to the crate specification.

5.2.2 Level C. The idler shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The idler is intended for use as a means of supporting and guiding 1-inch and 2-inch chain during the installation and recovery of tanker mooring legs made up of chain, chain connectors and fluked anchors.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Time frame required for submission of preproduction model and number of idlers required (see 3.2).
(c) When a preproduction pack is not required (see 5.1).
(d) Time frame required for submission of preproduction pack (see 5.1).
(e) Level of packing required (see 5.2).

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6.3 Preproduction model. Any changes or deviations of production idlers from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish idlers conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to pack and mark the idlers in accordance with this specification.

6.5 Supplementary quality assurance provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Definitions.

6.6.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of this specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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